# Traffic Flow Modeling

1. Counting Cars
   - flow and concentration
   - the continuity equation
2. Speed, Flow, and Concentration
   - the Greenshields model
   - flow as a function of concentration
3. Accelerating Traffic Flow
   - the transport equation
   - characteristic curves

MCS 472 Lecture 38
Industrial Math & Computation
Jan Verschelde, 17 April 2026

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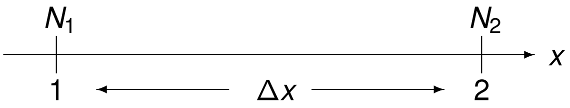
# counting cars

On two measure points (e.g., one mile apart), we count cars:




- $N_1$ is the number of cars counted at point 1,
- $N_2$ is the number of cars counted at point 2.

$$\Delta N = N_2 - N_1$$

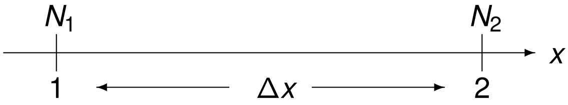
The direction of traffic moves from 1 to 2.

If $\Delta N < 0$, then $N_2 < N_1$, more cars at 1 than at 2,

which means that the concentration of cars between 1 and 2 increases.

# flow

On two measure points (e.g., one mile apart), we count cars:




Let $q$ be the number of cars per time unit $\Delta t$.

$$\begin{array}{rcl} q_2 & = & N_2/\Delta t \\ -\ q_1 & = & N_1/\Delta t \\ \hline \Delta q & = & \Delta N/\Delta t \end{array}$$

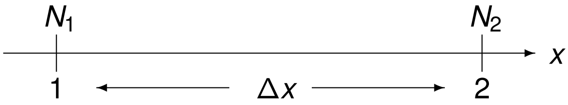
We have $\Delta N = \Delta q \Delta t$. As $\Delta t > 0$, $\Delta q$ has the same sign as $\Delta N$.

If $\Delta N < 0$, when the concentration of cars increases,

then $\Delta q < 0$ and the flow of cars decreases.

# concentration

On two measure points (e.g., one mile apart), we count cars:




Let $k$ be the number of cars per unit length $\Delta x$.

$$\begin{array}{rcl} k_1 & = & N_1/\Delta x \\ -\ k_2 & = & N_2/\Delta x \\ \hline \Delta k & = & (-\Delta N)/\Delta x \end{array}$$

If $\Delta N < 0$, then the flow of cars decreases
and the concentration increases, as $\Delta N < 0$ implies $\Delta k > 0$.

# Traffic Flow Modeling

# the continuity equation

If there is no exit nor entry between the measure points 1 and 2, then we have *continuity* and

$$\begin{aligned} \Delta N &= \Delta q \Delta t, \\ \Delta N &= -\Delta k \Delta x. \end{aligned}$$

Eliminating $\Delta N$ leads to $\Delta q \Delta N = -\Delta k \Delta x$, or equivalently:

$$\frac{\Delta q}{\Delta x} + \frac{\Delta k}{\Delta t} = 0.$$

Taking limits as $\Delta x \to 0$ and $\Delta t \to 0$, we have *the continuity equation*

$$\frac{\partial q}{\partial x}(x,t) + \frac{\partial k}{\partial t}(x,t) = 0.$$

Adding the change in flow to the change in concentration equals zero.

# on the units in the equation

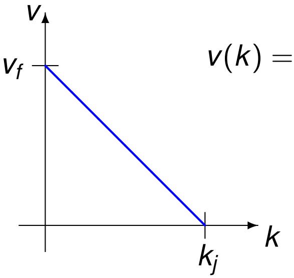
Exercise 1:
Let $\Delta x$ be one mile and $\Delta t$ be one hour.

Consider the equation
$$\frac{\Delta q}{\Delta x} + \frac{\Delta k}{\Delta t} = 0.$$
Verify that both terms in the equation have the same units, so the addition makes sense.

# Traffic Flow Modeling

# the Greenshields model

- The units of concentration $k$ are the number of cars per mile.
- The units of speed $v$ are the number of miles per hour.



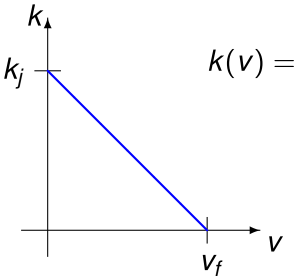

$$v(k) = v_f \left( 1 - \frac{k}{k_j} \right)$$

In the Greenshields model, the speed depends on the concentration.

- $v_f$ is the *free traffic speed*, when the road is free.
- $k_j$ is the *jam density*, $v = 0$ at a traffic jam.

# concentration in function of speed

$$v(k) = v_f \left( 1 - \frac{k}{k_j} \right) = v_f \left( \frac{k_j - k}{k_j} \right) \quad \Rightarrow \quad \begin{array}{l} k_j v = v_f(k_j - k) \\ k_j v - v_f k_j = -k v_f \end{array}$$



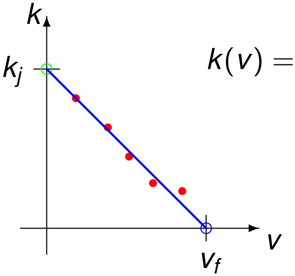

$$k(v) = k_j \left( 1 - \frac{v}{v_f} \right)$$

In the Greenshields model, the concentration depends on the speed.

- If $v = 0$, then traffic is jammed and $k = k_j$.
- If $v = v_f$, then there is no traffic and $k = 0$.

# experimental verification

While $v_f$ can be set to the speed limit,
the jam density $k_j$ is the intercept of a least squares fit.




$$k(v) = k_j \left( 1 - \frac{v}{v_f} \right)$$

The solid red dots represent observed data points.

# Traffic Flow Modeling

# flow as a function of concentration

Speed $v$ and concentration $k$ are linearly correlated.

Flow $q$ is expressed in the number of cars per hour.

Observe the following.

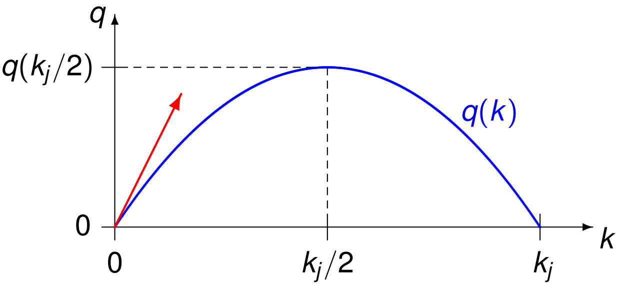
- If $k = 0$ (no cars per mile), then no cars per hour, $q = 0$.
- If $k = k_j$ (traffic is jammed), then no speed and no flow either.
- The maximum flow occurs for some $k$ in between zero and $k_j$.

Flow and concentration can thus not be correlated linearly.

Therefore, let us use a concave down parabola.

# a parabolic model for flow

$$q(k) = a_2 k^2 + a_1 k + a_0$$

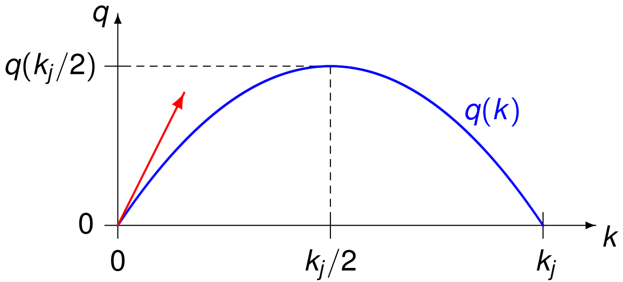



1. $q(0) = 0 \Rightarrow a_0 = 0$
2. $q'(k) = 2a_2 k + a_1 \Rightarrow a_1 = q'(0)$
3. $q''(k) = 2a_2 < 0$, concave down

*Where is $v_f$?*

# no flow at the traffic jam

$$q(k) = a_2 k^2 + q'(0)k = k\left(a_2 k + q'(0)\right)$$

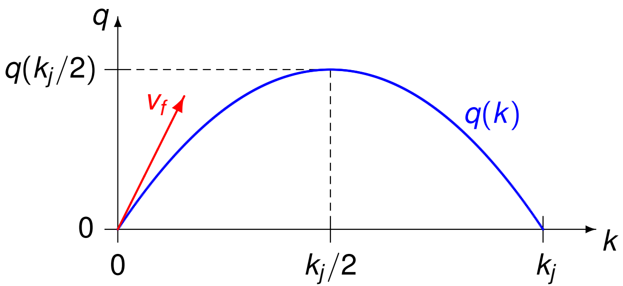



There is no flow at the traffic jam concentration, at $k = k_j$:

$$q(k_j) = \underbrace{k_j}_{\neq 0}\left(a_2 k_j + q'(0)\right) \quad \Rightarrow \quad a_2 = -\frac{q'(0)}{k_j}.$$

# applying the continuity equation

$$q(k) = k\left(-\frac{q'(0)}{k_j}k + q'(0)\right) = q'(0)k\left(1 - \frac{k}{k_j}\right)$$




By the continuity equation,

$$\frac{\Delta q}{\Delta x} + \frac{\Delta k}{\Delta t} = 0 \quad \Rightarrow \quad q(k) = v_f k\left(1 - \frac{k}{k_j}\right).$$

# on the units in the equation

Exercise 2:

Let $\Delta x$ be one mile and $\Delta t$ be one hour.

Consider the $\Rightarrow$ in the equation

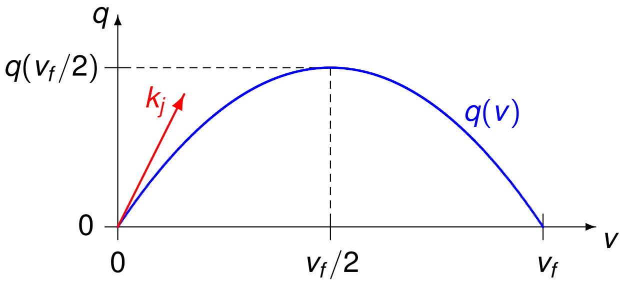
$$\frac{\Delta q}{\Delta x} + \frac{\Delta k}{\Delta t} = 0 \quad \Rightarrow \quad q(k) = v_f k \left( 1 - \frac{k}{k_j} \right).$$

Use the units to verify that the slope of the traffic flow equals indeed the free traffic speed at zero concentration.

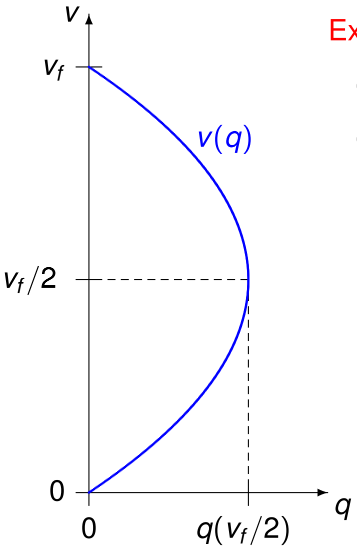
# flow as a function of speed

Because speed and concentration are linearly correlated as

$$k(v) = k_j \left( 1 - \frac{v}{v_f} \right) \quad \Rightarrow \quad q(v) = k_j v \left( 1 - \frac{v}{v_f} \right).$$



# speed as a function of flow



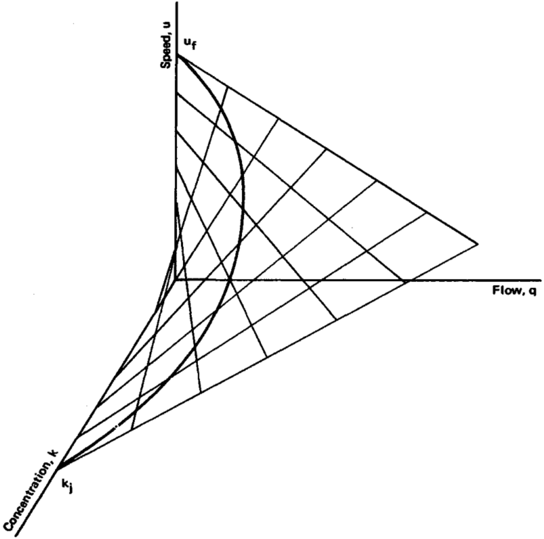

Exercise 3:

(1) Compute the formula for $v(q)$.

(2) Write a paragraph
to explain the picture.

# a 3D model

## Figure 4.1 in *Traffic Flow Theory* by D. L. Gerlough and M. J. Huber



# Traffic Flow Modeling

# the transport equation

The model relates speed $v$ to flow $q$ and concentration $k$ as

$$v = \frac{q}{k} \quad \text{with units} \quad \frac{\text{number of cars per hour}}{\text{number of cars per mile}} = \frac{\textit{mile}}{\textit{hour}}.$$

If in the continuity equation

$$\frac{\partial q}{\partial x} + \frac{\partial k}{\partial t} = 0$$

we replace $q$ by $k$ times $v$:

$$v\frac{\partial k}{\partial x} + \frac{\partial k}{\partial t} = 0,$$

then we obtain the transport equation with solution

$$k(x,t) = f(x - vt), \quad \text{where } f \text{ is the density profile at } t = 0.$$

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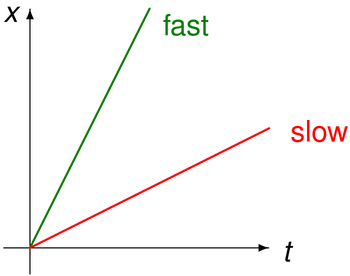
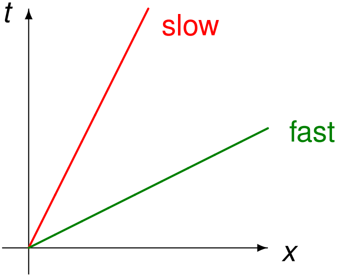
# characteristic curves

For some constant $C$, $x - vt = C$ defines *a characteristic curve*.




How much distance traveled in one hour?

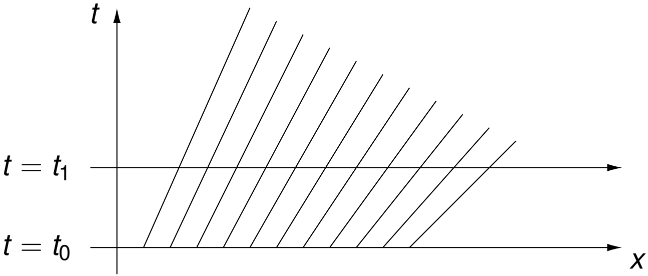



How much time to travel one mile?

# an accelerating platoon of cars

Consider a sequence of cars.

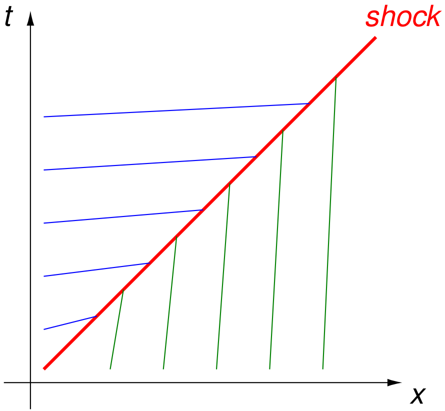
- At $t = t_0$, the spacing between all cars is the same.
- At $t > t_0$, the lead car accelerates.




At the horizontal cross sections, we see the position of each car.

Observe that no two lines meet so we have no collisions.

# a fast moving car meets a slow moving truck

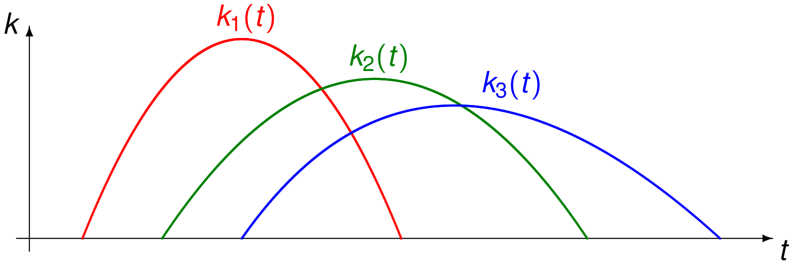



Exercise 4: Write one paragraph to interpret this picture.

# moving cars when stop light flips from red to green

The transport equation with diffusion: $v\dfrac{\partial k}{\partial x} + \dfrac{\partial k}{\partial t} = v\alpha^2 \dfrac{\partial^2 k}{\partial x^2}$

has a traveling concentration profile as solution:




1. $k_1(t)$ is the concentration at the red light,
2. $k_2(t)$ is the concentration at the green light,
3. $k_3(t)$ is the concentration after the green light.

# least squares fitting

The models are experimentally verified via least squares.

Proposal for a Topic of a Project:
Apply least squares fitting on observed traffic data.

Do the following:

- Collect data from an real highway.
- Fit the data for $k(v)$ and $q(v)$.
- Examine how realistic the models are.

In particular, can your model

- constructed with the data from one day,
- predict the traffic of the next day?

# summary and bibliography

We derived the continuity equation to study traffic flow and explored Greenshields model to relate flow, speed and concentration.

Traffic flow is covered in section 11.6 of our text book.


- Charles R. MacCluer:
  *Industrial Mathematics. Modeling in Industry, Science, and Government*. Prentice Hall, 2000.


An extensive study on traffic can be found in the report below.


- Daniel L. Gerlough and Matthew J. Huber:
  *Traffic Flow Theory. A Monograph*.
  Transportation Research Board Special Report 165, 1975.


The method of characteristic curves is covered in chapter 12 of


- Richard Haberman:
  *Applied Partial Differential Equations with Fourier Series and Boundary Value Problems*. Pearson, 2013.